Machine Learning
Assignment 8

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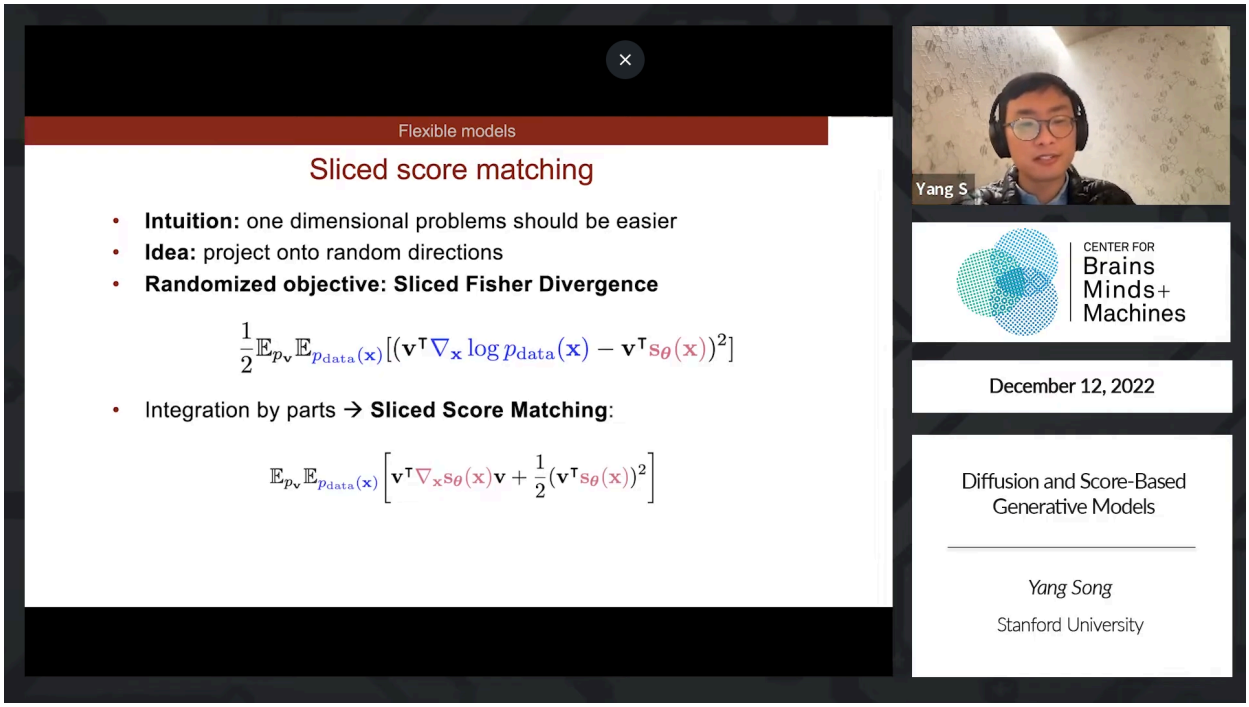
I. Written Assignment

- Show that the sliced score matching (SSM) loss can also be written as

$$L_{SSM} = \mathbb{E}_{x\sim p(x)}\mathbb{E}_{v\sim p(v)}[\|v^T S(x;\theta)\|^2 + 2v^T \nabla_x (v^T S(x;\theta))].$$

Ans.



First,

$$\mathbb{E}_{x\sim p(x)}[(v^T S(x;\theta))(v^T \nabla_x \log p(x))] = \int (v^T S(x;\theta))(v^T \nabla_x \log p(x))p(x)dx$$

And,

$$\nabla_x \log p(x) \nabla_x (\ln p(x)) = \frac{1}{p(x)} \nabla_x p(x)$$

we replace $\nabla_x \log p(x)$ with $\dfrac{\nabla_x p(x)}{p(x)}$. Then we have

$$\mathbb{E}_{x\sim p(x)}[(v^T S(x;\theta))(v^T \nabla_x \log p(x))] = \int (v^T S(x;\theta))(v^T \nabla_x p(x))dx$$

The following is integration by part：

$$\int (v^T S(x;\theta))(v^T \nabla_x p(x))dx = -\int p(x)(v^T \nabla_x (v^T S(x;\theta)))dx$$

So, we will have

$$\mathbb{E}_{x\sim p(x)}\mathbb{E}_{v\sim p(v)}\left[\left(v^T \nabla_x \log p(x) - v^T S(x;\theta)\right)^2\right]$$

$$= \mathbb{E}_{x\sim p(x)}\mathbb{E}_{v\sim p(v)}\left[v^T S(x;\theta)^2 - 2(v^T S(x;\theta)(v^T \nabla_x \log p(x))) + \left(v^T \nabla_x \log p(x)\right)^2\right]$$

$$= \mathbb{E}_{x\sim p(x)}\mathbb{E}_{v\sim p(v)}[\|v^T S(x;\theta)\|^2 + 2v^T \nabla_x (v^T S(x;\theta))]$$

Since the last term $v^T \nabla_x (v^T S(x;\theta))$ is constant, we omit it while optimizing. Then we have this result.

- Briefly explain SDE. A stochastic differential equation (SDE) is an ODE contaminated through a stochastic term. And it is of the form:

  $$dX_t = f(X_t, t) + G(X_t, t)dW_t$$

  where $f(X_t, t)$ is the drift(deterministic trend) and $G(X_t, t)dW_t$ is the diffusion term driven by Brownian motion W_t.

  The main method of solution is to find the PDF as a function of time using the equivalent Fokker-Planck equation(FPE). Since FPE is a deterministic PDE. It tells how the PDF evolves in time.

- **Unanswered Questions：**

  1. What is p represent in the below equation? $\frac{dp}{dt} = \frac{\sigma^2}{2} P_{xx}$
  2. How to implement Fokker-Planch Equation to solve SDE?
  3. What is the relation between them?
  4. How do we use Fokker-Planck Equation to solve SDEs with different PDFs, since we only discuss Gaussian distribution in class?

  Ans.

  p in the equation from question 1 denotes the probability of the particle being at x at time t in [1], [2].

  In [3], it tells us how to derive the solution of SDEs via Fokker-Planck equation. And the relation between SDEs and Fokker-Planck equation is also mentioned in [3]. The author uses Kramers-Moyal expansion to deduce Fokker-Planck equation from SDEs(In chapter 3 and 4).

  In [4], it approaches non-Gaussian normal diffusion via Fokker-Planck equation which tells us the step of solving the SDEs that are not Gaussian distribution.

II. Reference

[1] Wikipedia: Fokker-Planck Equation.

[2] Wikipedia: Stochastic differential equation.

[3] Hannes Risken. The Fokker-Planck Equation.

[4] Sumiyoshi Abe[1−4]. Fokker-Planck approach to non-Gaussian normal diffusion: Hierarchical dynamics for diffusing diffusivity.